The Effect of Correlated Experimental Uncertainties on Bayesian Inference of Quark–Gluon Plasma Properties

How correlated are the data?

Thiago S. Domingues, Nara Moraes, Matthew Luzum
USP – University of São Paulo

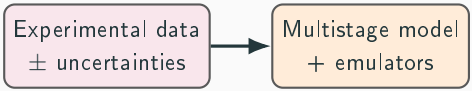
LASNPA 2026 – IPEN, São Paulo | [TODO: day] December 2026

Extracting QGP properties from data

Experimental data
$\pm$ uncertainties

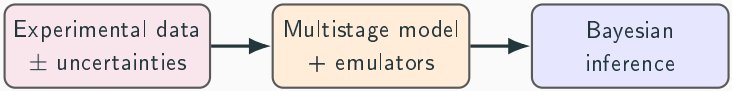
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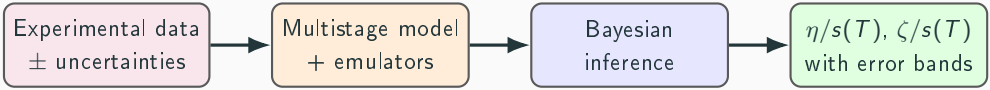
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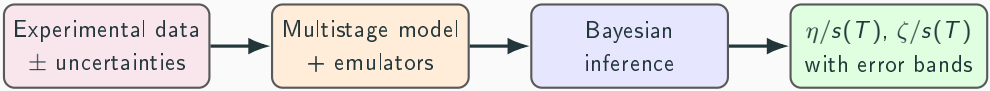
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Extracting QGP properties from data



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This talk

The width of the posterior depends on an input we do not have: the correlations between experimental uncertainties.

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Where the experimental uncertainty enters

$$P(\boldsymbol{\theta} \mid D) \propto \mathcal{L}(D \mid \boldsymbol{\theta})\, P(\boldsymbol{\theta}), \qquad \ln \mathcal{L} = -\frac{1}{2}\, \Delta y^{T}\, \Sigma^{-1}\, \Delta y - \frac{1}{2} \ln \det \Sigma + \text{const}$$

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- Standard practice: $\Sigma_{ij}^{\text{exp}} = \sigma_i^2\, \delta_{ij}$ (all points treated as independent).

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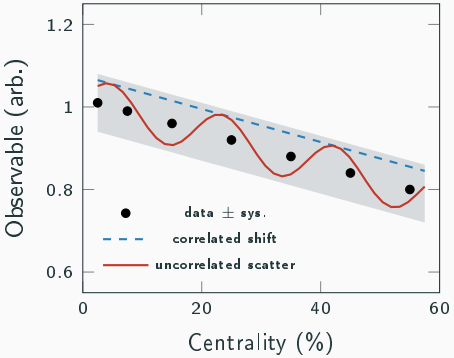
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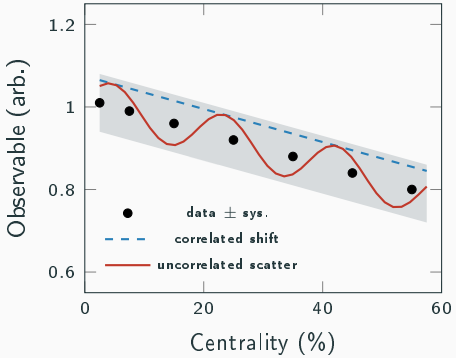
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Both curves have the same diagonal χ^2. Only one of them is a plausible systematic effect.

Why it matters: how many independent measurements?

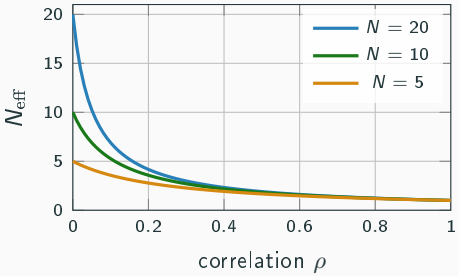
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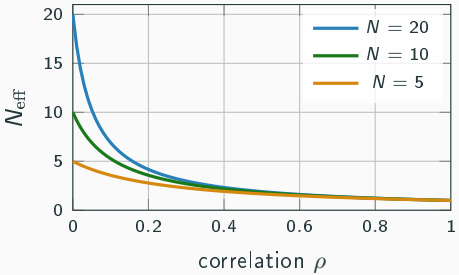


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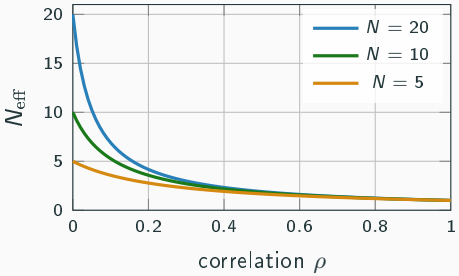


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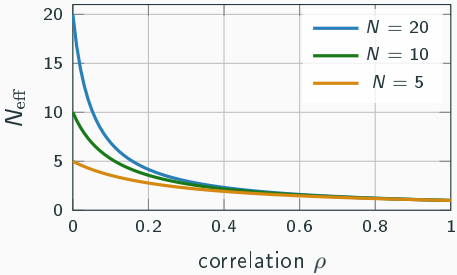
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Not just a rescaling of error bars

Correlations change which *directions* in data space are well measured: posteriors can broaden, narrow or shift.

A known warning: the viscosity posterior broadens



JETSCAPE Collaboration, Phys. Rev. C 103, 054904 (2021), App. D

A known warning: the viscosity posterior broadens

$\eta/s(T)$ and $\zeta/s(T)$ posteriors (Grad)
solid: diagonal $\Sigma^{\exp}$ dashed: correlated $\Sigma^{\exp}$
(JETSCAPE, PRC 103, 054904, Fig. 23)

jetscape_fig23_viscous_posterior.pdf

Same data, same model.

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- A simple correlated ansatz visibly **broadens** the η/s and ζ/s bands.
- The quoted uncertainty of QGP transport coefficients depends on an untested assumption.

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Two simple correlation models

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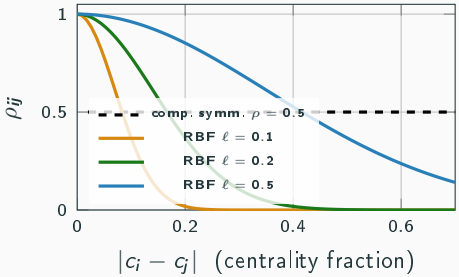
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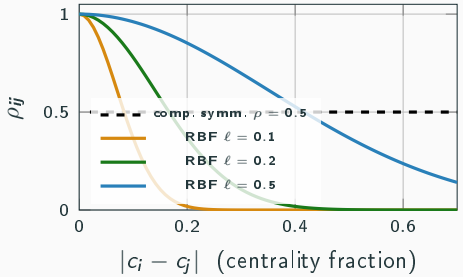
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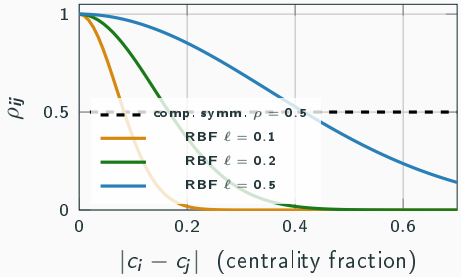
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Both are assumptions

Neither ρ nor ℓ is known. How much do the conclusions depend on them?

Setup

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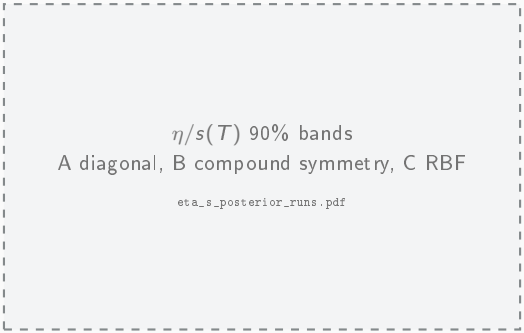
Run	$\Sigma^{\exp}$
A	diagonal (standard)
B	compound symmetry, $\rho =$ [TODO: values]
C	RBF, $\ell =$ [TODO: values]
D	informed by the data (GP)

Only $\Sigma^{\exp}$ changes between runs.

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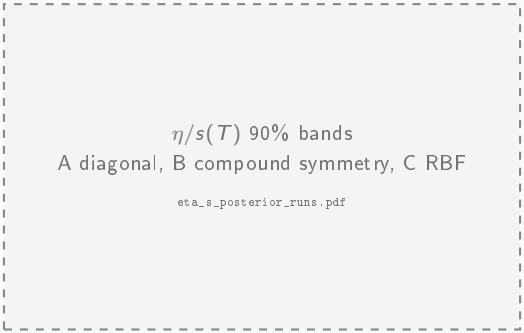
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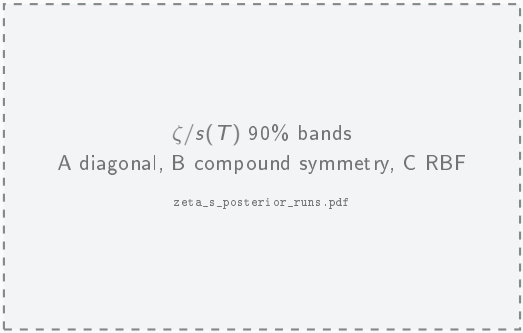


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$\eta/s(T)$ 90% bands
A diagonal, B compound symmetry, C RBF

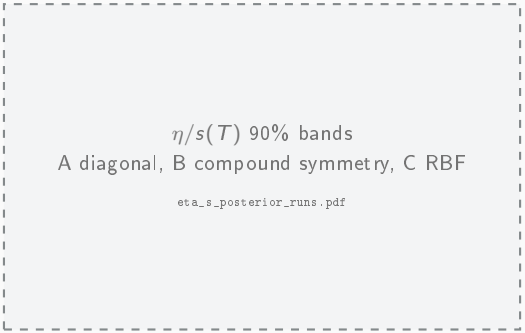


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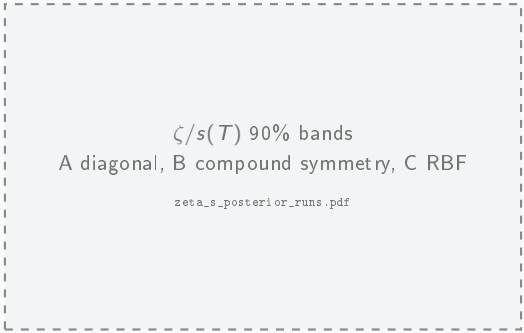


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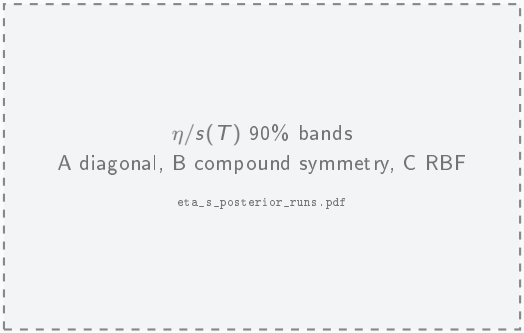
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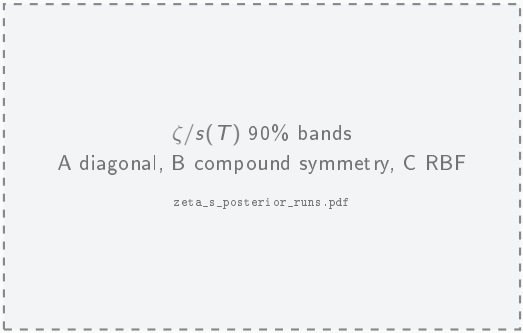
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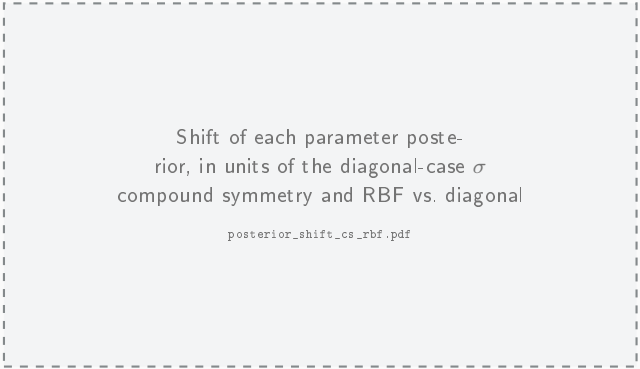


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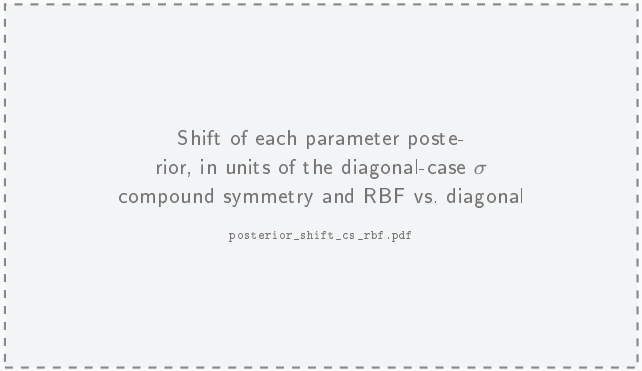


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- [TODO: result: trend with ρ and with ℓ]

Result II: the parameter posteriors shift



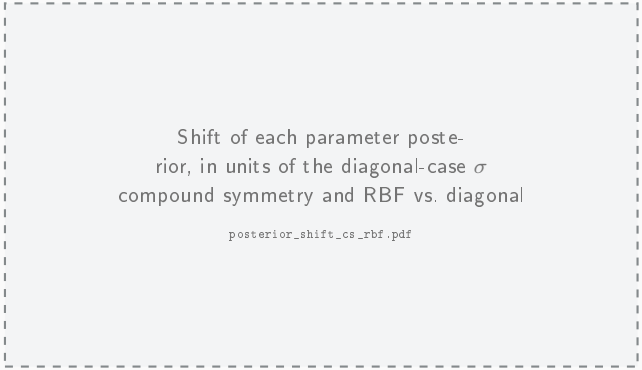
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Shift of each parameter posterior, in units of the diagonal-case σ
compound symmetry and RBF vs. diagonal

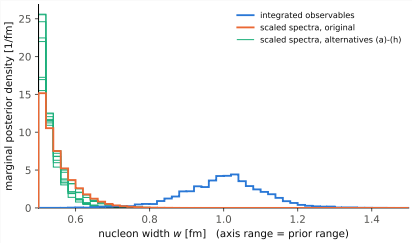
posterior_shift_cs_rbf.pdf

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- [TODO: which parameters shift most; setup of this study]

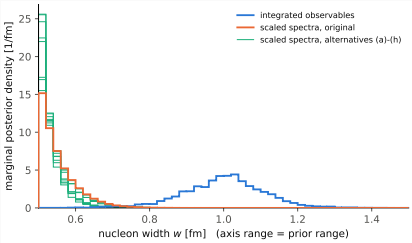
Case study: correlations do not always weaken a constraint



Scaled pion spectra $U(x_T)$ prefer $w \approx 0.5$ fm; integrated observables prefer $w \approx 1$ fm.

Scaled spectra: Domingues et al., arXiv:2602.22490. ALICE Pb–Pb 2.76 TeV π^+, 0–60%, 287 points.

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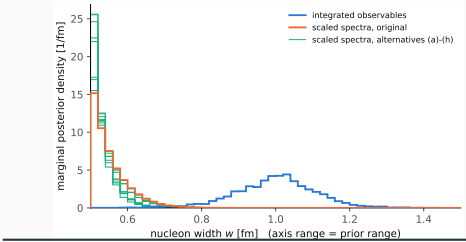


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Case study: correlations do not always weaken a constraint



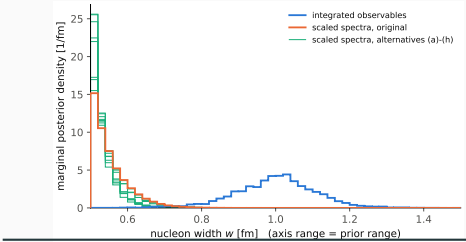
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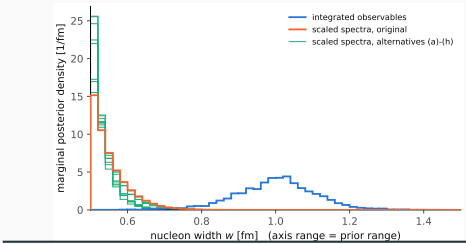
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Lesson

The effect has no fixed sign: it depends on how the observable is built.

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Can the data tell us? A Gaussian process on the measurements

For each observable, versus centrality c:

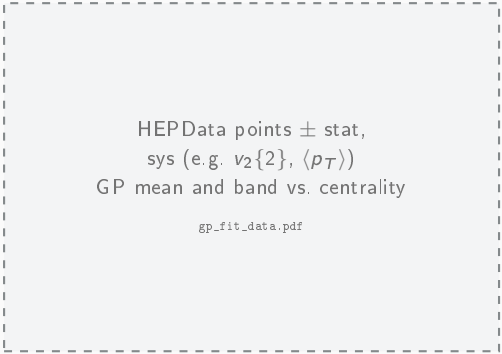
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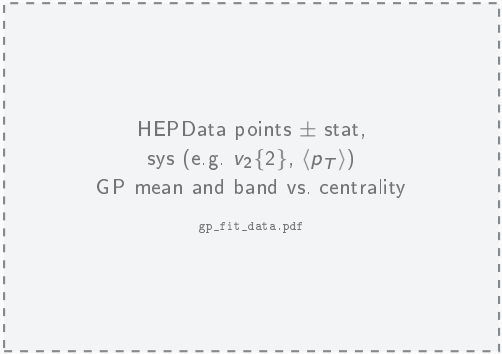


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HEPData points $\pm$ stat,
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GP mean and band vs. centrality

gp_fit_data.pdf

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The key signal: points scatter around a smooth curve **much less** than their systematic error bars allow.

First step: what correlation do the data support?

Pulls of HEPData points
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pull_distribution.pdf

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Why now

As statistical and emulator errors shrink, the unknown off-diagonal terms become a leading uncertainty in QGP property extraction.

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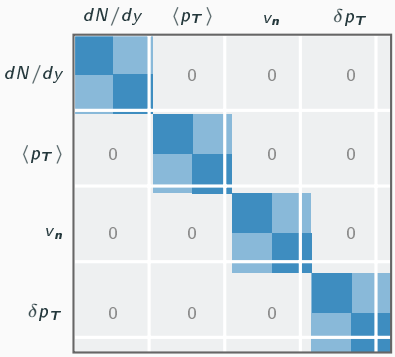
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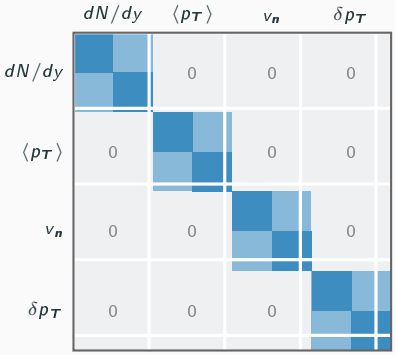
Backup

Backup: structure of the covariance matrix



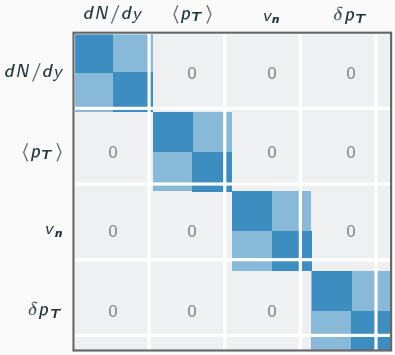
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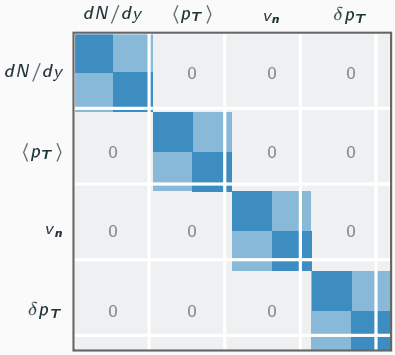
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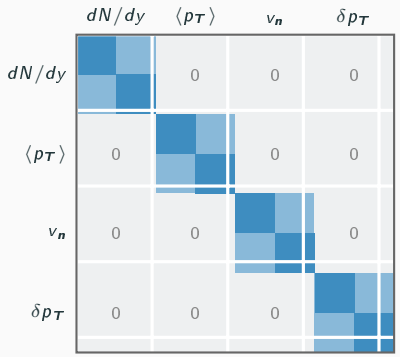


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The matrix stays positive semi-definite for any $\ell > 0$, $0 \leq \rho_0 \leq 1$.

Backup: what the data can and cannot tell us

Constrained

- The **uncorrelated** fraction $1 - \alpha$: bounded by the bin-to-bin scatter.
- Short-range correlation between neighbouring bins.

Ratio: $\sigma_R^2 / R^2 = \sigma_1^2/y_1^2 + \sigma_2^2/y_2^2 - 2\rho\,\sigma_1\sigma_2/(y_1 y_2)$ gives ρ directly.

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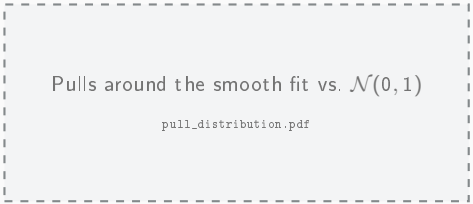
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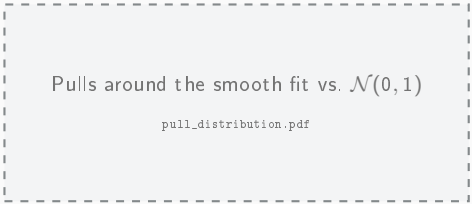
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Extra handles

- published uncertainties of **ratios** (species, energies, harmonics);
- the same observable from **different experiments**;
- systematics listed by source on HEPData.

Backup: scaled spectra, all alternative covariances

	$\sigma/\sigma_{\mathrm{orig}}$	χ^2_{ML} (287 pts)	$-2\Delta \ln L$ at $w = 1$	median w [fm]	90% interval [fm]	$P(w_{\mathrm{sp}} \geq w_{\mathrm{int}})$
Integrated observables	—	—	—	1.00	[0.81, 1.17]	—
Original: diagonal	**1.00**	**26**	**8.1**	**0.54**	**[0.50, 0.66]**	**0.0013**
(a) original budget, corr. modes	1.06	43	10.0	0.53	[0.50, 0.64]	0.0010
(b) uncorrelated, $\rho_c = 0$	0.72	450	9.2	0.54	[0.50, 0.65]	0.0014
(c) smooth, $\ell = 1$, $\rho_c = 0$	0.51	126	8.1	0.53	[0.50, 0.65]	0.0014
(d) smooth, $\ell = 1$, $\rho_c = 1$	0.51	837	16.8	0.52	[0.50, 0.60]	0.0005
(e) smooth, $\ell = 0.3$, $\rho_c = 1$	0.64	760	23.1	0.52	[0.50, 0.60]	0.0004
(f) coherent, $\rho_c = 1$	0.48	994	20.1	0.52	[0.50, 0.60]	0.0005
(g) norm. + tilt, $\rho_c = 0$	0.54	190	12.1	0.52	[0.50, 0.61]	0.0006
(h) norm. + tilt, $\rho_c = 1$	0.60	664	22.2	0.52	[0.50, 0.59]	0.0004

- ρ_c: correlation of the correlated part between centrality classes. Posteriors are bounded below by the prior edge $w = 0.5$ fm.
- With the original errors $\chi^2_{\mathrm{ML}}/287 = 0.09$: a sign that they are inflated.
- Caveats: emulator unchanged and used in extrapolation; one chain per alternative, w stable along each.

Backup: propagating uncertainties to the scaled spectra

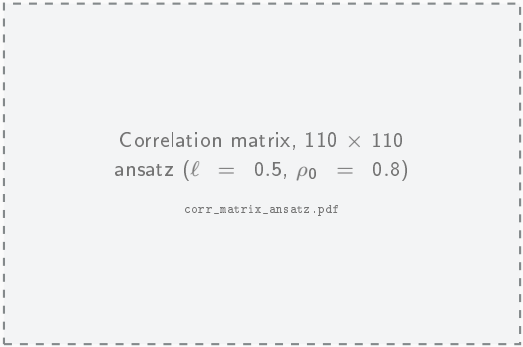
With $U(x_T) = (\langle p_T \rangle / N)\, \mathrm{d}N/\mathrm{d}p_T$ on a fixed x_T grid, to linear order:

$$\delta U_i = U_i \frac{\delta y_i}{y_i} - U_i \frac{\delta N}{N} + \left(U_i + x_i U_i'\right) \frac{\delta \langle p_T \rangle}{\langle p_T \rangle},$$

where δN and $\delta \langle p_T \rangle$ follow from the same fluctuations δy_j of the measured spectrum, plus the low-p_T extrapolation uncertainties.

- A common normalization change cancels in U, apart from the extrapolated part of the yield.
- The last term is the horizontal shift of the points when $\langle p_T \rangle$ changes.
- A non-linear Monte Carlo agrees with the linear propagation to 1–3%.

Backup: correlation matrices



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Correlation matrix, 110×110
ansatz ($\ell = 0.5$, $\rho_0 = 0.8$)

corr_matrix_ansatz.pdf

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- **[TODO: result: inferred α and ℓ per observable group; where they differ from the ansatz]**

Backup: which parameters are most sensitive?

90% interval of each of the 17 parameters
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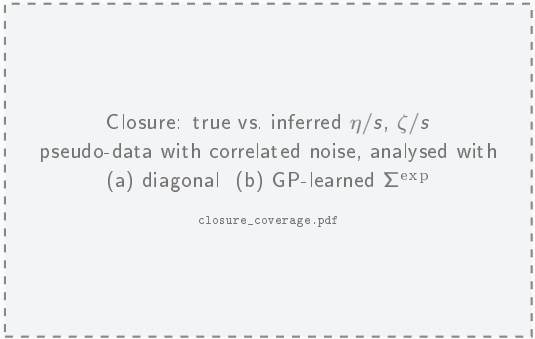
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Pseudo-data from the model at known $\boldsymbol{\theta}_{\mathrm{true}}$, with noise drawn from a **known correlated** $\sum^{\mathrm{exp}}$.

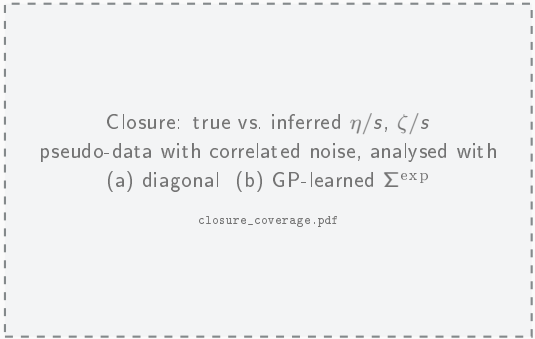
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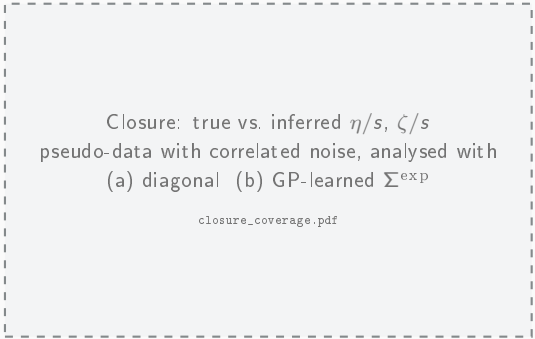
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- Diagonal analysis: does the 90% band contain the truth 90% of the time?

Backup: closure tests

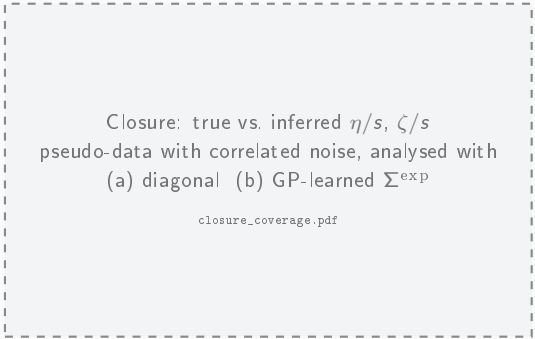


Pseudo-data from the model at known $\boldsymbol{\theta}_{\mathrm{true}}$, with noise drawn from a **known correlated** Σ^{exp}.

- Does the GP recover the injected (α, ℓ)?
- Diagonal analysis: does the 90% band contain the truth 90% of the time?
- GP-learned analysis: is coverage restored?

Backup: closure tests

Closure: true vs. inferred η/s, ζ/s
pseudo-data with correlated noise, analysed with
(a) diagonal (b) GP-learned Σ^{exp}



Pseudo-data from the model at known $\boldsymbol{\theta}_{\mathrm{true}}$, with noise drawn from a **known correlated** Σ^{exp}.

- Does the GP recover the injected (α, ℓ)?
- Diagonal analysis: does the 90% band contain the truth 90% of the time?
- GP-learned analysis: is coverage restored?
- [TODO: result]

Backup: full likelihood

$$\ln \mathcal{L}(\boldsymbol{\theta}, \ell, \rho_0) = -\frac{1}{2}\Delta y^T \Sigma^{-1} \Delta y - \frac{1}{2}\ln\det\Sigma - \frac{n}{2}\ln(2\pi),$$
$$\Delta y = y_{\mathrm{exp}} - y_{\mathrm{emu}}(\boldsymbol{\theta}), \qquad \Sigma = \Sigma^{\mathrm{exp}}(\ell, \rho_0) + \Sigma^{\mathrm{emu}}(\boldsymbol{\theta}).$$

- With fixed Σ^{exp} the determinant only matters through $\Sigma^{\mathrm{emu}}(\boldsymbol{\theta})$.
- With free (ℓ, ρ_0) it acts as an Occam penalty: stronger correlations shrink $\det\Sigma$ but make deviations orthogonal to the correlated direction more costly.

Backup: statistical versus systematic parts

A more careful decomposition keeps the statistical part diagonal:

$$\Sigma_{ij}^{\mathrm{exp}} = \left(\sigma_i^{\mathrm{stat}}\right)^2 \delta_{ij} + \rho_{ij}\, \sigma_i^{\mathrm{sys}}\, \sigma_j^{\mathrm{sys}}.$$

- Statistical errors of different bins are independent (different events).
- Exception: observables built from the same events (e.g. $v_2\{2\}$ and $v_3\{2\}$ in one centrality bin) have correlated statistical errors too.
- [TODO: state which convention is used in the results]

Backup: positive semi-definiteness

The ansatz is a product of valid kernels:

$$\rho_{(a,i),(b,j)} = \underbrace{R_{ab}}_{\text{observables}} \underbrace{\exp\left[-\frac{(c_i - c_j)^2}{\ell^2}\right]}_{\text{centrality}}, \qquad R_{ab} = \begin{cases} 1 & a = b \\ \rho_0 & a \neq b, \text{ same group} \\ 0 & \text{different groups} \end{cases}$$

- The squared-exponential kernel is positive semi-definite for any $\ell > 0$.
- R is block diagonal with eigenvalues $1 - \rho_0$ and $1 + (m-1)\rho_0$ (m observables in the group): positive for $0 \leq \rho_0 < 1$.
- Their product (Schur product theorem) is positive semi-definite.
- In practice: large ℓ makes Σ^{exp} nearly singular; Σ^{emu} and σ^{stat} regularize it.

Backup: two-point intuition

Two measurements with errors σ and correlation ρ. In the rotated variables

$$y_{\pm} = \frac{y_1 \pm y_2}{\sqrt{2}}: \qquad \mathrm{Var}(y_+) = \sigma^2(1+\rho), \qquad \mathrm{Var}(y_-) = \sigma^2(1-\rho).$$

- The **sum** (overall level) is known less precisely.
- The **difference** (slope, centrality dependence) is known more precisely.
- A parameter constrained by the level broadens; one constrained by the slope narrows.

Backup: alternative correlation models

- **Fully correlated normalization:** one nuisance shift per observable, $y \to (1+\epsilon)\, y$, $\epsilon \sim \mathcal{N}(0, \sigma_{\mathrm{norm}})$. Equivalent to a rank-one term in Σ^{exp}.
- **Exponential kernel:** $\rho_{ij} = \exp(-|c_i - c_j|/\ell)$, rougher than the Gaussian form.
- **Separate ℓ per group:** flow systematics need not share the length scale of yields.
- **Cross-group correlations:** e.g. yields and $\langle p_T \rangle$ share tracking efficiency and p_T extrapolation.

[TODO: keep only the variants actually tested; add comparison plot]



Backup: prior and posterior predictive

Observables: posterior predictive, diagonal Σ^{exp}

posterior_predictive_diag.pdf

Observables: posterior predictive, marginalized Σ^{exp}

posterior_predictive_corr.pdf

With correlated errors a good fit may sit coherently above or below the points: judge it with the full χ^2, not by eye.

Backup: full parameter posterior

Corner plot of all model parameters
diagonal (blue) vs. correlated/marginalized (orange)

corner_diag_vs_corr.pdf